Fisher Information and Efficiency

Math 742

2026-02-17

Introduction

Imagine that you are a detective and you are trying to find the location of a hidden treasure. Let's call the treasure θ. Some clues are large, clear and obvious. Others are small and barely visible. In order to decide if a clue provides a lot of information or a little bit of information, you take a measurement. The measurement gives you an indication of how much *information* is in a clue. **Fisher's information** does exactly that. It tells us whether the "clue" tells us a lot about θ or very little.

- High Fisher information $\implies$ the clue is very sharp/precise meaning your estimator can be very accurate.
- Low Fisher information $\implies$ the clue is blurry, even if you have a lot of data, your estimate of θ is uncertain.

We can even put a bound on how uncertain the measurement is (i.e. Cramér–Rao bound). This will give us a "feel" for why some parameters are easier to estimate than others.

In this lecture, we will try and answer this question:

"What is the lowest possible variance of an unbiased estimator?"

Score Function

We've seen already that the score function is defined as:

$$U(x|\theta) = \frac{\partial}{\partial\theta} \log f(X|\theta)$$

The score captures how sensitive the log-likelihood is to changes in θ.

Theorem: Let $f(x|\theta)$ be a pdf (or pmf) whose support does not depend on θ. Assume that we can interchange differentiation and integration. Then, $\mathbb{E}[U(\theta)] = 0$.

Proof:

By definition of expectation,

$$\mathbb{E}[U(\theta)] = \int \frac{\partial}{\partial\theta} \log f(x \mid \theta)\, f(x \mid \theta)\, dx.$$

Recall, the derivative of the log (with chain rule) is:

$$\frac{\partial}{\partial\theta} \log f(x \mid \theta) = \frac{1}{f(x \mid \theta)} \frac{\partial}{\partial\theta} f(x \mid \theta),$$

using that,

$$\mathbb{E}[U(\theta)] = \int \frac{1}{f(x \mid \theta)} \frac{\partial}{\partial\theta} f(x \mid \theta)\, f(x \mid \theta)\, dx = \int \frac{\partial}{\partial\theta} f(x \mid \theta)\, dx.$$

Since the support of $f(x \mid \theta)$ does not depend on θ and differentiation may be interchanged with integration,

$$\mathbb{E}[U(\theta)] = \frac{\partial}{\partial\theta} \int f(x \mid \theta)\, dx = \frac{\partial}{\partial\theta}(1) = 0.$$

Fisher Information

Definition 1: For a single observation,

$$I(\theta) = \mathbb{E}[(U(\theta))^2] = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta} \log f(X|\theta)\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(X|\theta)\right]$$

Note: In order for this statement to be true, the same "regularity" conditions must hold. (i.e. (1) the support does not depend on θ and (2) we assume that interchanging differentiation and integration is fine.)

Fisher Information:

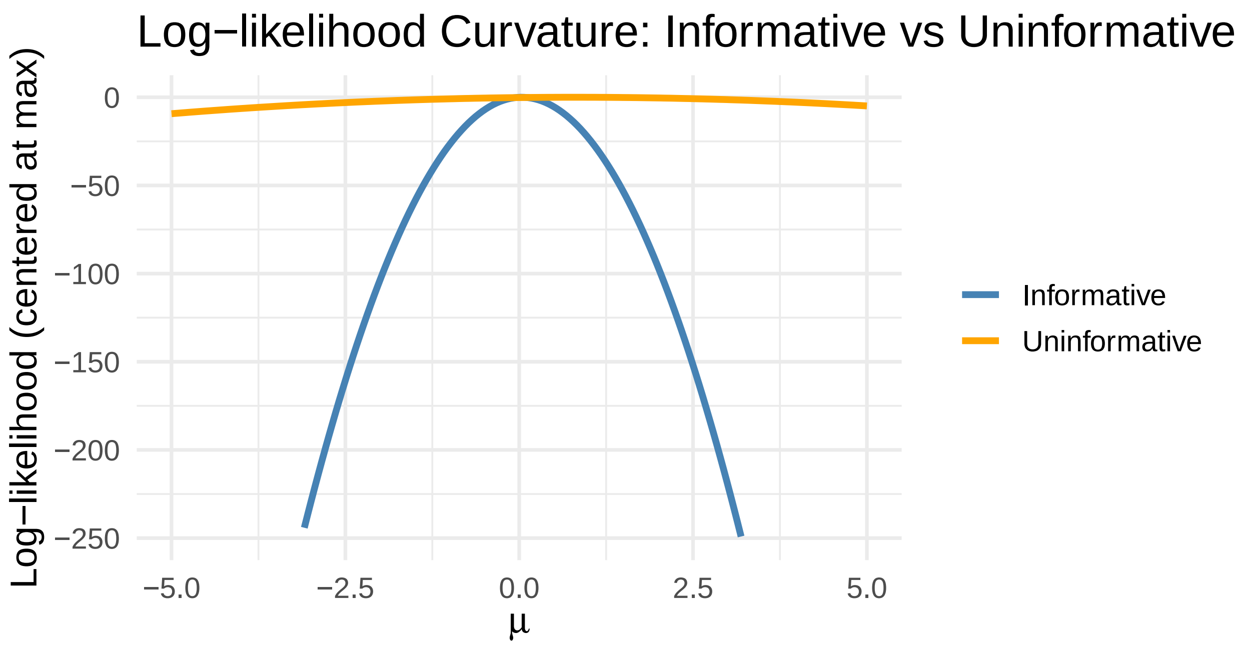
- The score function measures how sensitive the log-likelihood is to θ.
- The Fisher information measures how much "signal" your data carries about θ.
- Since $\mathbb{E}[U(\theta)] = 0$, $\mathbb{V}[U(\theta)] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(X|\theta)\right] = -\mathbb{E}[\ell''(\theta)]$ $(\ell(\theta) = \log f(X|\theta))$.

The last bullet links the variance of the score function to the curvature of the likelihood. Think of it this way, the log-likelihood curve $(\ell(\theta))$ is the landscape.

- The first derivative $U(\theta) = \ell'(\theta)$ is the slope of the hill.
- The second derivative $\ell''(\theta)$ is the curvature. How sharply the hill bends.

$$\boxed{\begin{array}{c} \rightarrow \text{“Curvature” of the log-likelihood around the true parameter.} \leftarrow \\ \text{High curvature} \Rightarrow \text{data is informative} \Rightarrow \text{lower variance achievable.} \\ \text{Low curvature} \Rightarrow \text{likelihood is flat} \Rightarrow \text{more uncertainty.} \end{array}}$$

Example: Informative vs. Non-informative information.



Both sample size and noise level affect curvature; this plot exaggerates the contrast by changing both. The "informative" curve uses large n and small variance, whereas the "uniformative" curve uses small n and large variance.

Definition 2: The Fisher information for an iid sample of size n,

$$I_n(\theta) = nI(\theta)$$

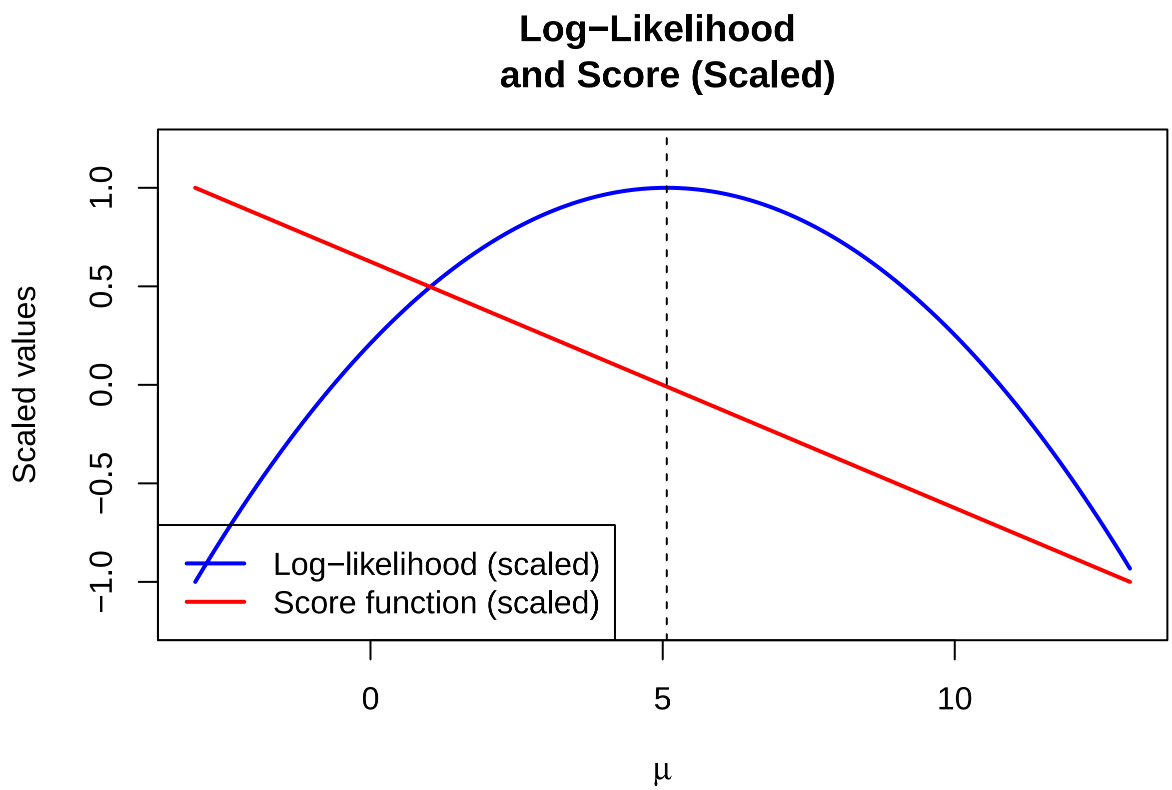
Example: Normal with known variance.

Consider an iid sample $X_1, X_2, ..., X_n \sim N(\mu, \sigma^2)$, where σ^2 is known.

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$$

$$\text{Score: } U(\mu) = \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum(x_i - \mu)$$

$$\text{Fisher Info: } I(\mu) = \frac{n}{\sigma^2}$$



- The blue curve shows the log-likelihood with a peak at the MLE estimate of μ (i.e. $\bar{x}$).
- The red curve shows the score function (i.e. $\ell'(\mu)$). It is equal to zero at the the MLE.
- The Fisher information equals $n/\sigma^2 = 50/4 = 12.5$ (i.e. it's constant and doesn't depend on μ so it does not appear on the plot.

Note: in order to get everything on the same axis I had to scale the functions appropriately. Scaling in no way affects interpretation.

Examples

$N(\mu, \sigma^2)$, μ unknown, $\sigma^2 < \infty$ unknown

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(X_i - \mu)^2$$

$$U(\mu) = \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum(X_i - \mu)$$

$$U(\sigma^2) = \frac{\partial \ell}{\partial \sigma^2} = \frac{1}{2(\sigma^2)^2}\left[\sum(X_i - \mu)^2 - n\sigma^2\right]$$

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2} \implies I_{\mu,\mu} = -\mathbb{E}\left[\frac{-n}{\sigma^2}\right] = \frac{n}{\sigma^2}$$

$$\frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} = \frac{-1}{(\sigma^2)^2}\sum(X_i - \mu)$$

$$-\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \mu \partial \sigma^2}\right] = 0 \implies I_{\mu,\sigma^2} = 0 \text{ (evaluated at MLE=}\bar{X}\text{)}$$

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2} = -\frac{\sum(X_i - \mu)^2}{(\sigma^2)^3} + \frac{n}{2(\sigma^2)^2}$$

$$\implies I_{\sigma^2\sigma^2} = -\mathbb{E}\left[-\frac{\sum(X_i - \mu)^2}{(\sigma^2)^3} + \frac{n}{2(\sigma^2)^2}\right] = \frac{n\sigma^2}{(\sigma^2)^3} - \frac{n}{2(\sigma^2)^2} = \frac{2n - n}{2(\sigma^2)^2} = \frac{n}{2(\sigma^2)^2}$$

$$\text{Fisher Information Matrix} = \text{Fisher Information Matrix} = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2(\sigma^2)^2} \end{bmatrix}$$

Note: We compute the second derivatives and then take expectations. In some cases (like , the second derivative is already constant, so the expectation has no effect.

Binomial(m, p)

Let $X_1, X_2, ..., X_n$ be an iid sample from Binomial(m, p).

$$\ell(p) = \left(\sum X_i\right)\log p + \left(nm - \sum X_i\right)\log(1 - p)$$

$$U(p) = \frac{d}{dp}\ell(p) = \frac{\sum X_i}{p} - \frac{nm - \sum X_i}{1 - p} = \frac{\sum X_i - nmp}{p(1 - p)}$$

$$\frac{d^2}{dp^2}\ell(p) = -\frac{\sum X_i}{p^2} - \frac{nm - \sum X_i}{(1 - p)^2}$$

$$I(p) = -\mathbb{E}\left[-\frac{\sum X_i}{p^2} - \frac{nm - \sum X_i}{(1 - p)^2}\right]$$

$$= \frac{nmp}{p^2} + \frac{nm(1 - p)}{(1 - p)^2} = \frac{nm}{p(1 - p)}$$

Poisson(λ)

$$\ell(\lambda) = \sum \big(X_i \log \lambda - \lambda - \log(X_i!)\big) = \left(\sum_{i=1}^{n} X_i\right) \log \lambda - n\lambda - \sum \log(X_i!)$$

$$U(\lambda) = \frac{\sum X_i}{\lambda} - n$$

$$I(\lambda) = -\mathbb{E}\left[\frac{d^2}{d\lambda^2}\ell(\lambda)\right]$$

$$I(\lambda) = \frac{n}{\lambda}$$

$$-\mathbb{E}\left[-\frac{\sum X_i}{\lambda^2}\right] = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$$

Summary:

Distr	MLE	Fisher Info	Efficient?
Bernoulli(p)	$\hat{p} = \bar{X}$	$I(p) = \frac{n}{p(1-p)}$	Yes
Binomial(m, p)	$\hat{p} = \frac{\sum X_i}{mn}$	$I(p) = \frac{mn}{p(1-p)}$	Yes
Poisson(λ)	$\hat{\lambda} = \bar{X}$	$I(\lambda) = \frac{n}{\lambda}$	Yes
Normal($\mu, 1$)	$\hat{\mu} = \bar{X}$	$I(\mu) = n$	Yes
Normal(μ, σ^2) — estimating μ	$\hat{\mu} = \bar{X}$	$I(\mu) = \frac{n}{\sigma^2}$	Yes
Normal(μ, σ^2) — estimating σ^2	$\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$	$I(\sigma^2) = \frac{n}{2\sigma^4}$	No

Note: Efficiency here refers to finite-sample efficiency among unbiased estimators (CRLB). The MLE of σ^2 is asymptotically efficient but not unbiased therefore technically not efficient.

Cramér–Rao Lower Bound (CRLB)

The CRLB provides a lower bound on the variance of any unbiased estimator.

Let $X_1, X_2, ..., X_n$ be an iid sample from a distribution with PDF $f(x|\theta)$. Let $\hat{\theta}$ be an unbiased estimate of $\theta \implies \mathbb{E}(\hat{\theta}) = \theta$.

The score function is defined as:

$$U(\theta) = \frac{\partial}{\partial\theta}\ell(\theta) = \frac{\partial}{\partial\theta}\log L(\theta)$$

And the Fisher information:

$$I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ell(\theta)\right]$$

The Cramér–Rao inequality is thus,

$$\mathbb{V}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

$$\implies \text{no unbiased estimator can have a variance smaller than } 1/I(\theta)$$

In order for this to be true we must have:

1. An unbiased estimator.
2. The density function $f(x|\theta)$ satisfies the regularity conditions.

- The support of $f(x|\theta)$ does not depend on θ
- Differentiation under the integral sign is allowed
- $I(\theta) < \infty$

Proof (Sketch, Cramér–Rao inequality):

Assume $\hat{\theta}$ is an unbiased estimator of θ, so that

$$\mathbb{E}(\hat{\theta}) = \int \hat{\theta}(x)\, f(x \mid \theta)\, dx = \theta.$$

Differentiating both sides with respect to θ and interchanging differentiation and integration,

$$\frac{d}{d\theta}\mathbb{E}(\hat{\theta}) = \int \hat{\theta}(x)\, \frac{\partial}{\partial\theta} f(x \mid \theta)\, dx = 1.$$

Using

$$\frac{\partial}{\partial\theta} f(x \mid \theta) = f(x \mid \theta)\, \frac{\partial}{\partial\theta} \log f(x \mid \theta),$$

we obtain

$$\mathbb{E}\left[\hat{\theta}\, \frac{\partial}{\partial\theta} \log f(X \mid \theta)\right] = 1.$$

Since $\mathbb{E}[U(\theta)] = 0$, this is equivalent to

$$\mathbb{E}\left[(\hat{\theta} - \theta)\, U(\theta)\right] = 1.$$

Applying the Cauchy–Schwarz inequality,

$$1 \leq \sqrt{\mathbb{E}\left[(\hat{\theta} - \theta)^2\right]}\, \sqrt{\mathbb{E}[U(\theta)^2]},$$

and hence

$$\mathbb{V}(\hat{\theta}) \geq \frac{1}{\mathbb{E}[U(\theta)^2]} = \frac{1}{I(\theta)},$$

where the final equality follows from the definition of Fisher information.

Example: Normal distribution with known variance (σ^2).

- $X_1, X_2, ..., X_n$ iid ($N(\mu, \sigma^2)$)
- Fisher information: $I(\mu) = n/\sigma^2$
- CRLB: $\mathbb{V}(\hat{\mu}) \geq \sigma^2/n$
- Since $\mathbb{V}(\bar{X}) = \sigma^2/n$, the sample mean achieves the Cramér–Rao lower bound and is therefore an efficient estimator.

Example: Poisson(λ)

- Fisher information: $I(\lambda) = n/\lambda$
- CRLB: $\mathbb{V}(\hat{\lambda}) \geq \lambda/n$
- Since $\bar{X}$ is unbiased for λ and $\mathbb{V}(\bar{X}) = \lambda/n$, it achieves the Cramér–Rao lower bound and is therefore efficient.

Efficiency

Efficiency tells us how small the variance could "theoretically" be.

Definition 3: The efficiency of an unbiased estimator is the ratio of the CRLB to the variance. If it achieves the CRLB the efficiency equals 1 and we say that it is an **efficient** estimator.

$$\text{efficiency}(\hat{\theta}) = \frac{\text{CRLB}}{\mathbb{V}(\hat{\theta})} = \frac{1/I(\theta)}{\mathbb{V}(\hat{\theta})} \leq 1$$

- If the ratio $\frac{1/I(\theta)}{\mathbb{V}(\hat{\theta})}$ equals 1, the estimator is efficient.
- If the $\frac{1/I(\theta)}{\mathbb{V}(\hat{\theta})}$ less than 1, the estimator is inefficient.

Unbaisedness $\Rightarrow$ The classic CRLB only applies to unbiased estimators
Efficiency is traditionally defined relative to the CRLB, so its's inherently a concept for unbiased estimators.

Definition 4: An estimator is **asymptotically efficient** if it achieves efficiency in the limit as $n \to \infty$.

Many estimators (especially MLEs) do not achieve the CRLB for finite samples, but they do achieve it asymptotically.

Example: Let $X_1, X_2, ..., X_n \sim N(\mu, \sigma^2)$

Let's look at the MLE for σ^2. We should earlier that,

$$\sigma^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

$$\mathbb{E}[\sigma^2_{MLE}] = \frac{n-1}{n}\sigma^2$$

So the MLE is biased.

$$\mathbb{V}(\sigma^2_{MLE}) = \frac{2\sigma^4(n-1)}{n^2}$$

We also showed that,

$$I(\sigma^2) = \frac{n}{2\sigma^4}$$

This makes the CRLB equal to $1/I(\sigma^2) = \frac{2\sigma^4}{n}$ Since the variance doesn't equal the lower bound it the MLE for σ^2 is not efficient, not to mention it's biased.

However, as $n \to \infty$

$$\frac{2\sigma^4(n-1)}{n^2} \to \frac{2\sigma^4}{n}$$

Also, since $\mathbb{E}[\sigma^2_{MLE}] = \frac{n-1}{n}\sigma^2$, $\text{Bias}(\sigma^2_{MLE}) = -\frac{\sigma^2}{n} \to 0$ as $n \to \infty$

$\to$ So the MLE of σ^2 is asymptotically efficient but not efficient.

Definition 5: The relative efficiency between estimators is defined as:

$$\text{relative efficiency} = \frac{\mathbb{V}(\hat{\theta}_2)}{\mathbb{V}(\hat{\theta}_1)}$$

This can be useful when neither estimator achieves the lower bound.

Implications for Data Scientists

1. Fisher Information tells you:

- How much data you need.
- How precise your estimate can be.
- Why some parameters are harder to estimate.

2. Connect to experimental design:

- More informative measurements reduce required sample size.

Modern Relevance

- **Model selection:** AIC $= -2\ell + 2k \approx$ deviance + penalty on number of parameters. AIC arises from a second-order Taylor expansion of the log-likelihood using Fisher information; the resulting penalty equals 2 per parameter, reflecting the dimension of the parameter space.
- **Experimental design:** Maximize expected Fisher info (e.g., optimal dosing in trials)
- **Deep learning:** Hessian $\approx$ observed info $\to$ second-order optimizers (Newton) use curvature for faster convergence
- **Uncertainty quantification:** Profile likelihood intervals use $I(\theta)$ for non-asymptotic CIs

Summary

- Score function $\to$ derivative of log-likelihood
- Fisher Information $\to$ variance of score
- CRLB $\to$ best achievable variance
- MLE $\to$ asymptotically hits the CRLB $\to$ asymptotically efficient

Takeaway – Fisher Information is Your Data's "Signal Strength"

- **Fisher information** $I(\theta)$ quantifies how much your data "knows" about θ.

  High $I(\theta) \to$ sharp likelihood $\to$ precise estimators (low Cramér-Rao bound).

  Low $I(\theta) \to$ flat likelihood $\to$ even the best unbiased estimator has high variance.

  MLE is asymptotically efficient: $\mathbb{V}(\hat{\theta}_{MLE}) \approx 1/(nI(\theta)) \to$ it achieves the Cramér-Rao lower bound for large n.

- In data science: Use it to compare models (AIC/BIC) or design experiments (maximize expected $I(\theta)$).

Distribution / Parameter	Estimator	Bias	Variance	CRLB	Achieves CRLB?
Bernoulli(p)	$\hat{p} = \bar{X}$	0	$\frac{p(1-p)}{n}$	$\frac{p(1-p)}{n}$	Yes
Binomial(m, p)	$\hat{p} = \frac{1}{mn}\sum X_i$	0	$\frac{p(1-p)}{mn}$	$\frac{p(1-p)}{mn}$	Yes
Poisson(λ)	$\hat{\lambda} = \bar{X}$	0	$\frac{\lambda}{n}$	$\frac{\lambda}{n}$	Yes
Normal($\mu, 1$)	$\hat{\mu} = \bar{X}$	0	$\frac{1}{n}$	$\frac{1}{n}$	Yes
Normal(μ, σ^2) — estimating μ	$\hat{\mu} = \bar{X}$	0	$\frac{\sigma^2}{n}$	$\frac{\sigma^2}{n}$	Yes
Normal(μ, σ^2) — estimating σ^2	$\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$	$-\frac{\sigma^2}{n}$	$\frac{2\sigma^4}{n}$	$\frac{2\sigma^4}{n}$	No (biased MLE)

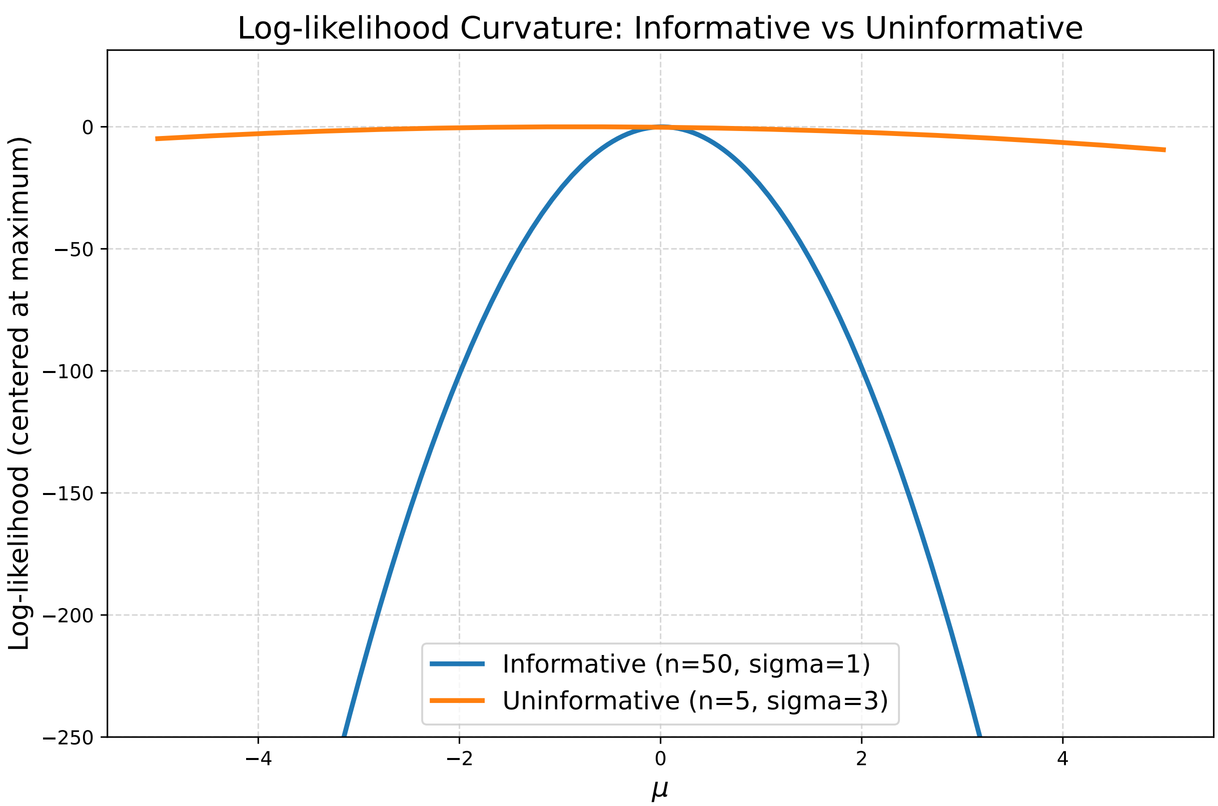
Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Set seed for exact reproducibility
np.random.seed(123)

# -------------------------------------------------
# 1. Informative data: large n, small variance
# -------------------------------------------------
n_info     = 50
sigma_info = 1.0
x_info     = np.random.normal(loc=0, scale=sigma_info, size=n_info)

# -------------------------------------------------
# 2. Uninformative data: small n, large variance
# -------------------------------------------------
n_uninfo     = 5
sigma_uninfo = 3.0
x_uninfo     = np.random.normal(loc=0, scale=sigma_uninfo, size=n_uninfo)

# -------------------------------------------------
# 3. Log-likelihood for Normal(mu, sigma^2), sigma known
# -------------------------------------------------
def loglik(mu, x, sigma):
    n = len(x)
    return (
        -0.5 * n * np.log(2 * np.pi * sigma**2)
        - np.sum((x - mu)**2) / (2 * sigma**2)
    )

# Grid of mu values
mu_grid = np.linspace(-5, 5, 200)

# Compute log-likelihoods
ll_info   = np.array([loglik(mu, x_info,   sigma_info)   for mu in mu_grid])
ll_uninfo = np.array([loglik(mu, x_uninfo, sigma_uninfo) for mu in mu_grid])

# Center at the maximum
ll_info   -= ll_info.max()
ll_uninfo -= ll_uninfo.max()

# -------------------------------------------------
```

```
# 4. Plot (matplotlib only)
# --------------------------------------------------
plt.figure(figsize=(9, 6))

plt.plot(mu_grid, ll_info,   linewidth=2.5, label="Informative (n=50, sigma=1)")
plt.plot(mu_grid, ll_uninfo, linewidth=2.5, label="Uninformative (n=5, sigma=3)")

plt.title("Log-likelihood Curvature: Informative vs Uninformative", fontsize=16)
plt.xlabel(r"$\mu$", fontsize=14)
plt.ylabel("Log-likelihood (centered at maximum)", fontsize=14)

plt.ylim(-250, None)
```

```
## (-250.0, 31.415937525294385)
```

```
plt.legend(fontsize=13, frameon=True)
plt.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```




```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(123)

# Parameters
n       = 50
```

```python
sigma   = 2.0
mu_true = 5.0

# Generate data
x      = np.random.normal(loc=mu_true, scale=sigma, size=n)
xbar    = np.mean(x)  # MLE = sample mean

# Grid of mu values
mu_vals = np.linspace(mu_true - 4*sigma, mu_true + 4*sigma, 400)

# Log-likelihood function
def loglik(mu):
    return -(n/2)*np.log(2*np.pi*sigma**2) - np.sum((x - mu)**2) / (2*sigma**2)

# Score function (derivative of log-likelihood)
def score(mu):
    return (1/sigma**2) * np.sum(x - mu)

# Theoretical Fisher Information (constant for location family)
fisher_info = n / sigma**2   # = 50 / 4 = 12.5

# Compute values
ll_vals    = np.array([loglik(mu)  for mu in mu_vals])
score_vals = np.array([score(mu)   for mu in mu_vals])

# Scaling function: rescale any vector to [-1, 1]
def scale_to_unit(v):
    v_min, v_max = v.min(), v.max()
    return (v - v_min) / (v_max - v_min) * 2 - 1

ll_scaled    = scale_to_unit(ll_vals)
score_scaled = scale_to_unit(score_vals)

# Fisher info is constant → manually place it at a visible level (like in your R code)
info_scaled = np.full_like(mu_vals, 0.7)

# ------------------------------
# Plot (ggplot2-style with matplotlib)
# ------------------------------
plt.figure(figsize=(10, 6))

plt.plot(mu_vals, ll_scaled,    color='steelblue',   linewidth=2.5, label='Log-likelihood (scaled)')
plt.plot(mu_vals, score_scaled, color='crimson',     linewidth=2.5, label='Score function (scaled)')


# Vertical line at MLE
plt.axvline(x=xbar, color='black', linestyle='--', linewidth=1.5, label=f'MLE = xbar approx {xbar:.2f}'

# Labels and title
plt.xlabel(r'$\mu$', fontsize=14)
plt.ylabel('Scaled values', fontsize=14)
plt.title('Log-Likelihood and \n Score Function (scaled)',
          fontsize=15, pad=20)
```

```
plt.ylim(-1.2, 1.2)
```

```
## (-1.2, 1.2)
```

```
plt.xlim(mu_vals.min(), mu_vals.max())
```

```
## (-3.0, 13.0)
```

```
# Legend
plt.legend(loc='lower left', frameon=True, fontsize=11)

# Clean look
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```